KAI WANG

Ph.D. Candidate in Economics & Finance

@ kai.wang@fsv.cuni.cz +420 774-654-115 Prague, Czech Republic
Institute of Economic Studies, Charles University



PROFILE

A Ph.D. candidate in Economics and Finance at Charles University, focusing on Digital Commodities and Financial Econometrics. Interested in applying machine learning and high-frequency econometrics to study risk transmission, factor structures, and contagion between centralized and decentralized trading platforms.

WORKING PAPERS & RESEARCH PROJECTS

Native Token as a Factor for Chain-Specific Tokens: A Factor Model and CAPM Analysis

Working paper

2024 – Present

- Examines whether platform-native tokens (e.g. ETH, BNB) span the risk premia of chain-specific tokens using CAPM-style and multifactor panel regressions.
- Builds a large token–day panel with mixed-frequency volatility and liquidity factors (MIDAS-style) and robust Driscoll–Kraay inference.

Who Leads on Which Horizon? A Multi-Layer Contagion Map Between DEX and CEX Platform Tokens

Project in progress

2025 – Present

- Constructs a high-frequency connectedness map between DEX and CEX platform tokens across intraday and multi-day horizons.
- Combines TVP–VAR and frequency-domain FEVD spillover measures to distinguish short- and long-horizon leadership and stress episodes.

ACADEMIC POSITIONS

Ph.D. Candidate & Seminar TA

Institute of Economic Studies, Charles University

Sep 2024 – Present Prague, Czech Republic

- **Econometrics I (JEB109); Econometrics II (JEB110)**
- **Corporate Finance (JEM034); Statistics (JEB034)**

Researcher

Institute of Information Theory and Automation (UTIA), Czech Academy of Sciences

Sep 2023 – Present Prague, Czech Republic

- Contributed to empirical finance projects on crypto-asset and energy markets using advanced econometric and time-series methods.

INDUSTRY EXPERIENCE

Corporate Due Diligence Analyst (Part-time)

Merck Sharp & Dohme (MSD)

Jan 2024 – Present Prague, Czech Republic

Risk Management Analyst (Contractor)

Industrial and Commercial Bank of China (ICBC)

May 2023 – Dec 2023 Prague, Czech Republic

Assistant to Vice President

Winland Group

Mar 2020 – Aug 2021 Beijing, China

EDUCATION

Ph.D. in Economics & Finance

Charles University

Sep 2024 – Present Prague, Czech Republic

Erasmus Exchange

Johannes Gutenberg University

Sep 2022 -Mar 2023 Mainz, Germany

M.Sc. in Economics

University of Economics,Prague (VŠE)

Sep 2021 -Jan 2024 Prague, Czech Republic

GPA: 1.86

B.A. in Business English

Jilin International Studies University

Sep 2016 -Jun 2020 Changchun, China

Avg. Score:81.43

METHODS & TOOLS

- **Programming:** R, Python, Stata, VBA.
- **Machine learning (basic):** scikit-learn, tree-based models, regularised regression; strong interest in extending to deep learning for time-series and order-book data.
- **Data & visualisation:** API-based collection and cleaning of high-frequency crypto and financial data (tick/1–5 min), feature construction, and visualisation in R and Python.

LANGUAGES

- Chinese (native), English (fluent)
- German (A1.1), French (A1).

CERTIFICATES

- GRE General Test: 335/340 (Quantitative 169/170).